Yi-Ting Lai

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EDUCATION

National Yang Ming Chiao Tung University Hsinchu, ROC
MS Computer Science, *Advisor: Yi-Ping You* GPA: 3.92/4.3 Sep 2023–Aug 2026

- Thesis: Improving Intra-Warp Cache Utilization on GPUs through Compiler-Guided Exploitation of Texture Hardware
- Courses: Advanced Compiler, Physical Design Automation, Accelerator Architectures for Machine Learning

National Taipei University of Technology Taipei, ROC
BS Electronic Engineering GPA: 3.96/4.0, **Rank: 1/116** Jun 2023

- **1st Place**, In-Campus Calculus Competition (245 participants, 2019)

EXPERIENCE

Advanced Semiconductor Engineering, Inc. Kaohsiung, ROC
IT Intern Jul–Aug 2022

- Automated defect detection using **YOLOv5**.
- Visualized factory production and downtime metrics by building data dashboards with **Apache Superset**.

TEACHING ASSISTANTSHIPS

NYCU Parallel Programming Sep–Dec 2024, Feb–Jun 2025
NYCU Compiler Design, **Excellent Teaching Assistant Award** Feb–Jun 2024
NYCU Introduction to Compiler Design Sep 2023–Jan 2024
NTUT Object-Oriented Programming Feb–Jun 2021

PROJECTS

AutoTex | C++ Master Thesis
An LLVM-based compiler framework for GPU memory-layout optimization.

- Reduced fetched sectors by 38% at L1, 15% at L2, and 21% at DRAM across 17 kernels, achieving 2.4× speedup, by developing **LLVM passes** that detect access/layout mismatches and map data to **texture memory**.

VitaminC | C++ 2023–2025
A C compiler targeting **QBE** and **LLVM IR**.

- Implemented key C features, including functions, arrays, pointers, loops, conditionals, structs, and unions.

Wav2Letter FPGA Accelerator | Verilog 2025

- Accelerated ASR inference by 15×, as measured by throughput, by designing a **systolic array** architecture.

Concentration Judgment System | Python 2021–2022
A student concentration analysis system for online courses.

- Achieved real-time face and posture detection using **Dlib** and **OpenCV**.
- Implemented GUI and client communication with **PyQt5** and **WebSocket**.

SKILLS

Programming: C/C++, CUDA, Python, Bash
Software & Tools: LLVM, Git, NVIDIA Nsight Compute, Make/CMake, GDB, Docker, Linux
Language: Mandarin Chinese (Native), English (Professional Proficiency, TOEIC 875), Japanese (Basic)

PUBLICATIONS

1. Yi-Ting Lai, Meng-Shiuan Pan. "*A Concentration Judgment System for Online Courses: Design and Implementation.*" VTS Asia Pacific Wireless Communications Symposium (APWCS), 2023.